MIL-C-83443/1A
30 May 1975
SUPERSEDING
MIL-C-83443/1
22 April 1974

MILITARY SPECIFICATION SHEET
CELL, SOLAR, SILICON, SILVER CONTACT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the solar cell described herein shall consist of this document and the issue in effect of Specification MIL-C-83443.

REQUIREMENTS:

Cell material: Single crystal, boron-doped, P-type silicon.
Resistivity: Nominally 2 ohm-cm and not lower than 1 ohm-cm.
Contact material: Principally silver with small quantities of titanium and palladium permitted. An overlay of solder may be used if requested by the procuring activity.
Electrical output:

Solar intensity:	135 mW/cm^2 maximum
Cell temperature:	28.0°C minimum
Cell voltage:	0.450 $\pm$0.002 V

Minimum cell output current at cell voltage:

	2.0 x 2.0 cm	2.0 x 4.1 cm	2.0 x 6.2 cm
Group 1	132 mA	271 mA	409 mA
Group 2	134 mA	275 mA	415 mA
Group 3	136 mA	279 mA	422 mA
Group 4	138 mA	283 mA	428 mA
Group 5	140 mA	287 mA	434 mA

A minimum acceptable average output of any shipping lot shall be based upon the above grouping and shall not be less than group 3.

FSC 6116

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:08Z
Check the source to verify that this is the current version before use.

Electrical output after cover glass installation: Average increase not less than 1 mA/cm^2.

Particle radiation resistance:

Electrons

Energy:	1 Mev
Fluence:	3×10^{14} e/cm^2 front side
Flux:	5×10^{10} to 1×10^{12} $/cm^2$ per second without heat sink

The cell shall not degrade to less than 80 percent of the original average minimum power requirement.

Radiometric properties: Solar absorptance = 0.84 maximum with cover glass installed.

QUALITY ASSURANCE PROVISIONS

Electrical output after cover glass installation:

Tab material for welding test 4.5.5.5: Silver with less than 1 percent impurity.

Tab material for soldering test, 4.5.5.4: Tinned copper stranded wire.

Cover glass material: Fused silica or CeO doped microsheet

Size: 150µm thick, covering the entire active area

Blue-reflective coating: 350 nanometers cuton

Anti-reflective coating: MgF_2

Cover glass adhesive: Dow Corning 93-500 or equivalent

Cell cleaning fluid: Water and alcohol.

Custodian:
Army -
Navy -
Air Force - 11

Preparing activity:
Air Force - 11

Project No. 6116-0002-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:08Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will - erwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:08Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

AFLC/MMGE/STANDARDIZATION
WRIGHT-PATTERSON AFB OH 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:08Z
Check the source to verify that this is the current version before use.